Ten Layers from Tail Absorption to the Annular Weighted-Prefix Frontier

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Abstract

RH-292–RH-301 audit the last analytic determinant leaf after the projection-free spectral tail. Tail absorption shortens the missing bridge from slope four to the sharp mass-and-cap slope $1/\log(10/7)$. We derive the exact uniform coefficient-rate threshold, upgrade coefficientwise convergence to a weighted bridge on an unspecified slow clock, and prove that unweighted agreement even on the exact clock is insufficient. The existing Gaussian orbit-tube clock is strictly too short. At the natural endpoint-resolution clock, the shortened bridge crosses one counterloop alias and the slope-four cut crosses two. The square-root parity term blocks separate absolute majorants. Zero-padded root transport gives a sharp conditional head rate, while annular H^∞ or H^2 convergence would directly close the coefficient norm without resolving roots. None of these actual quantitative inputs is presently verified. Both typed ledgers remain four-of-five and Gates A–E remain open.

1 The shortened bridge

Let $R = 7/5$, $q = 1/2$, and

$$m_\sigma = \lceil 4L_\sigma \rceil, \qquad h_\sigma = \lceil a_* L_\sigma \rceil, \qquad a_* = \frac{1}{\log(10/7)}, \qquad L_\sigma = \log(1/\sigma).$$

RH-292 proves

$$P_\sigma(m_\sigma) \le P_\sigma(h_\sigma) + S_\sigma(h_\sigma) + T_\sigma(h_\sigma),$$

where the last two terms tend to zero and $S_\sigma(h_\sigma) = O(L_\sigma^{-1})$. Thus the shortest bridge clock certified by the current sharp mass-and-cap envelope has slope $a_* = 2.803673252\ldots$, not four. RH-293 shows that a uniform coefficient error $O(\sigma^\beta)$ has the sharp information-class threshold

$$\beta_* = a_* \log R = 0.9433582098747317\ldots,$$

with logarithmic decay at equality.

RH-294 strengthens the fixed-order noisy/counterloop limits to a weighted full-trace constituent on some selected $h_\sigma \to \infty$. It gives no lower bound on that clock. RH-295 proves why a rate is indispensable: one escaping coefficient spike can have a vanishing maximum on the prescribed window and a divergent weighted budget.

2 Physical clock firewalls

The existing interior orbit-tube proof, when required uniformly over every order in the prefix, has ceiling

$$a_{\text{loc}} = \frac{1}{\log \lambda} = 1.930709419\ldots < a_*.$$

RH-296 therefore proves an empty overlap between the current localization architecture and every mass-and-cap tail-admissible bridge clock. This is a proof-method stop, not actual trace nonconvergence.

If the counterloop period is tied to the intrinsic endpoint rank $k_\sigma = L_\sigma/(2\log\lambda) + O(1)$, RH-297 gives

$$2k_\sigma < h_\sigma < 4k_\sigma < m_\sigma < 6k_\sigma.$$

The minimal bridge contains one exact alias and the slope-four bridge contains two. Since $\beta R = 1.271273304\ldots > 1$, their isolated shell weights grow; a positive theorem must retain their cancellation or renormalize them.

RH-298 combines raw fixed-order localization with $1 + \lambda_-(\sigma) = C_*\sqrt{\sigma} + o(\sqrt{\sigma})$. It obtains

$$\frac{c^H_{\sigma,n} - c^H_n}{\sqrt{\sigma}} \to (-1)^n n C_* r_H^{-n}.$$

The separate absolute parity budget grows on the minimal clock with exponent $a_* \log(R/r_H) - 1/2 = 0.899008185\ldots$. The combined extracted trace may still cancel; only the separate-majorant architecture is excluded.

3 Two admissible quantitative interfaces

For finite multisets padded by zeros, RH-299 proves

$$|\sum x_j^n - \sum y_j^n| \le nB^{n-1} d_1^0(X,Y).$$

At $B = \beta$ and the minimal clock, the disk-bounded Lipschitz route is guaranteed by a root cost decaying faster than $\sigma^{0.6729348509\ldots}$. The radial family makes this information-class threshold sharp. No such noisy modulus-head matching is known, and this route would supply only the head constituent.

RH-300 gives the direct aggregate alternative. If

$$g_\sigma(z) = \sum_{n\ge 2} \frac{(\tau_{\sigma,n} - a_n)z^n}{n}$$

is holomorphic on a neighborhood of $|z| \le \rho$ for $\rho > R$ and has a controlled annular norm, then

$$P_\sigma^\infty(R) \le \|g_\sigma\|_{H^\infty(\rho)} \frac{(R/\rho)^2}{1 - R/\rho},$$

or

$$P_\sigma^\infty(R) \le \|g_\sigma\|_{H^2(\rho)} \frac{(R/\rho)^2}{\sqrt{1-(R/\rho)^2}}.$$

The certified target radius leaves the strict annulus $1.4 < \rho < 1.426787483864073\ldots$. An endpoint counterexample shows why the H^2 radius gap matters.

4 Typed frontier

branch	head	bridge	tail	target	boundary	score
noisy modulus spectrum	1	0	1	1	1	4
graded monodromy counterloop	1	1	0	1	1	4

The rate-free weighted full-trace bridge does not change the bridge bit because it is not synchronized to h_σ. The actual head transport and direct annular aggregate are also absent. Hence the weighted cross-branch glue bit is false and the complete count is zero.

Proposition 4.1 (next executable leaf)**.** *The preferred reopening input is vanishing annular H^∞ or H^2 norm for the actual complement-to-anchor logarithmic mismatch. The typed alternative requires both a parity-renormalized, alias-inclusive full-trace estimate and an actual modulus-head transport estimate on the minimal clock.*

Proof. RH-300 turns either annular norm directly into the absolute weighted prefix, and RH-292 absorbs the remaining slab and tails. Otherwise RH-288 gives $P \leq E + D$; RH-296–RH-299 identify the precise clock, parity, alias, and root rate obligations of those two terms. □

Remark 4.2. No Gate-A determinant identification has been completed. Gates B–E remain false/open. Nothing in this batch constructs a Hilbert–Polya operator, identifies Riemann zeros, proves a von Mangoldt trace formula, proves a completed-zeta divisor identity, or implies RH.